

Ricordiamo che abbiamo dimostrato:

$\Sigma_0: AX = 0$ sistema lineare omogeneo su K in n variabili.

$S_0 \subseteq K^n$ insieme delle soluzioni.

$$S_0 = \left\{ (x_1, \dots, x_n) \in K^n \mid A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0 \right\}$$

S_0 è un sottosp. vett. di K^n e $\dim S_0 = n - \text{rang}(A)$

Esercizio:

$$\Sigma_0: \begin{cases} x_4 + x_2 - x_3 + 2x_4 + x_5 = 0 \\ 2x_4 + x_2 + x_3 + x_4 - x_5 = 0 \\ x_4 + 2x_3 - x_4 - 2x_5 = 0 \end{cases}$$

Determinare S_0 e una sua base

$$A = \begin{pmatrix} \boxed{1} & 1 & -1 & 2 & 1 \\ \textcircled{2} & 1 & 1 & 1 & -1 \\ \textcircled{1} & 0 & 2 & -1 & -2 \end{pmatrix} \quad \begin{array}{l} \underline{q}^1 \rightarrow \underline{q}^1 - 2\underline{q}^2 \\ \underline{q}^3 \rightarrow \underline{q}^3 - \underline{q}^1 \end{array} \quad \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 0 & \boxed{-1} & 3 & -3 & -3 \\ 0 & \textcircled{-1} & 3 & -3 & -3 \end{pmatrix}$$

$$\underline{q}^3 \rightarrow \underline{q}^3 - \underline{q}^2 \quad \begin{pmatrix} \boxed{1} & \textcircled{1} & -1 & 2 & 1 \\ 0 & \boxed{-1} & 3 & -3 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \underline{q}_2 \rightarrow -\underline{q}_2 \\ \underline{q}_1 \rightarrow \underline{q}_1 - \underline{q}_2 \end{array} \rightarrow \begin{pmatrix} 1 & 1 & -1 & 2 & 1 \\ 0 & 1 & -3 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 2 & -1 & -2 \\ 0 & 1 & -3 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{cases} x_1 = -2x_3 + x_4 + 2x_5 \\ x_2 = 3x_3 - 3x_4 - 3x_5 \end{cases}$$

$$S_0 = \{ (-2x_3 + x_4 + 2x_5, 3x_3 - 3x_4 - 3x_5, x_3, x_4, x_5) \mid x_3, x_4, x_5 \} \subseteq \mathbb{R}^5$$

$\dim S_0 = 3$

$$(-2x_3, 3x_3, x_3, 0, 0) + (x_4, -3x_4, 0, x_4, 0) + (2x_5, -3x_5, 0, 0, x_5) =$$

$$= x_3(-2, 3, 1, 0, 0) + x_4(1, -3, 0, 1, 0) + x_5(2, -3, 0, 0, 1)$$

$$\Rightarrow_{T=} \{ (-2, 3, 1, 0, 0), (1, -3, 0, 1, 0), (2, -3, 0, 0, 1) \} \text{ è un nat. d. gen. di } S_0$$

$|T| = 3 = \dim S_0$

$\Rightarrow T$ è una base di S_0

Oss:

$$\begin{pmatrix} -2 & 3 & 1 & 0 & 0 \\ 1 & -3 & 0 & 1 & 0 \\ 2 & -3 & 0 & 0 & 1 \end{pmatrix}$$

3 colonne lin. indep.

Esercizio

$$\Sigma_0: \begin{cases} x_2 - x_3 + 2x_4 + x_5 = 0 \\ x_4 + 2x_2 + x_3 - x_4 + 2x_5 = 0 \\ x_4 + 3x_2 + x_4 + 3x_5 = 0 \\ 2x_1 + 3x_2 + 3x_3 + x_4 - x_5 = 0 \end{cases}$$

5 variabili; $k = \mathbb{R}$

Determinare $\mathcal{B}_0$ e una base

$$A = \begin{pmatrix} 0 & 1 & -1 & 2 & 1 \\ 1 & 2 & 1 & -1 & 2 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & 1 & -1 \end{pmatrix} \quad \underline{q}^1 \leftrightarrow \underline{q}^2 \quad \begin{pmatrix} \boxed{1} & 2 & 1 & -1 & 2 \\ 0 & 1 & -1 & 2 & 1 \\ \textcircled{1} & 3 & 0 & 1 & 3 \\ \textcircled{2} & 3 & 3 & 1 & -1 \end{pmatrix}$$

$$\begin{aligned} \underline{q}^3 &\rightarrow \underline{q}^3 - \underline{q}^1 \\ \underline{q}^4 &\rightarrow \underline{q}^4 - 2\underline{q}^1 \end{aligned} \quad \begin{pmatrix} 1 & 2 & 1 & -1 & 2 \\ 0 & \boxed{1} & -1 & 2 & 1 \\ 0 & \textcircled{1} & -1 & 2 & 1 \\ 0 & \textcircled{-1} & 1 & 3 & -5 \end{pmatrix} \quad \begin{aligned} \underline{q}^3 &\rightarrow \underline{q}^3 - \underline{q}^2 \\ \underline{q}^4 &\rightarrow \underline{q}^4 + \underline{q}^2 \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & -1 & 2 \\ 0 & 1 & -1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & -4 \end{pmatrix} \quad \underline{q}^4 \rightarrow \underline{q}^3 \quad \begin{pmatrix} \boxed{1} & 2 & 1 & -1 & 2 \\ 0 & \boxed{1} & -1 & 2 & 1 \\ 0 & 0 & 0 & \boxed{5} & -4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \underline{q}^3 \rightarrow \frac{1}{5} \underline{q}^3$$

$$\begin{pmatrix} 1 & 2 & 1 & \textcircled{-1} & 2 \\ 0 & 1 & -1 & \textcircled{2} & 1 \\ 0 & 0 & 0 & \boxed{1} & -\frac{4}{5} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{aligned} \underline{q}^2 &\rightarrow \underline{q}^2 - 2\underline{q}^3 \\ \underline{q}^1 &\rightarrow \underline{q}^1 + \underline{q}^3 \end{aligned} \quad \begin{pmatrix} 1 & \textcircled{2} & 1 & 0 & \frac{6}{5} \\ 0 & \boxed{1} & -1 & 0 & \frac{13}{5} \\ 0 & 0 & 0 & 1 & -\frac{4}{5} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\underline{q}^1 \rightarrow \underline{q}^1 - 2\underline{q}^2 \quad \begin{pmatrix} \boxed{1} & 0 & 3 & 0 & -\frac{4}{5} \\ 0 & \boxed{1} & -1 & 0 & \frac{13}{5} \\ 0 & 0 & 0 & \boxed{1} & -\frac{4}{5} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{aligned} x_3 & & x_4 \\ & & & & \\ & & & & \end{aligned} \quad \frac{6}{5} - \frac{26}{5} = -\frac{20}{5} = -4$$

$$\dim \mathcal{B}_0 = 2$$

$$\begin{cases} x_4 = -3x_3 + 4x_5 \\ x_2 = x_3 - \frac{13}{5}x_5 \\ x_1 = \frac{4}{5}x_5 \end{cases}$$

$$\mathcal{B}_0 = \left\{ (-3x_3 + 4x_5, x_3 - \frac{13}{5}x_5, x_3, \frac{4}{5}x_5, x_5) \mid x_3, x_5 \in \mathbb{R} \right\}$$

||

$$(-3x_3, x_3, x_3, 0, 0) + (4x_5, -\frac{13}{5}x_5, 0, \frac{4}{5}x_5, x_5) =$$

$$= x_3(-3, 1, 1, 0, 0) + x_5(4, -\frac{13}{5}, 0, \frac{4}{5}, 1)$$

$$\Rightarrow \mathcal{B}_0 = \mathcal{L} \left((-3, 1, 1, 0, 0), (4, -\frac{13}{5}, 0, \frac{4}{5}, 1) \right)$$

$$\text{base} = \left\{ (-3, 1, 1, 0, 0), (4, -\frac{13}{5}, 0, \frac{4}{5}, 1) \right\}$$

$$\begin{pmatrix} -3 & 1 & \boxed{1} & 0 & \boxed{0} \\ 4 & -\frac{13}{5} & 0 & \frac{4}{5} & \boxed{1} \end{pmatrix}$$

$$\begin{pmatrix} 4 & -13/5 & 0 & 4/5 & 1 \end{pmatrix}$$

2 colonne lin. indep.

Teorema. Sia $W \subseteq K^m$ sottosp. vett. numerico.

Esiste $\Sigma_0: AX=0$ tale che $S_0 = W$.

DIM Sia $B = (w_1, \dots, w_h)$ una base ordinata di W , $\dim W = h$.

$$w_1 = (a_{11}^1, a_{12}^1, \dots, a_{1m}^1) \in W \subseteq K^m$$

$$W = \mathcal{L}(B)$$

$$\vdots$$

$$w_h = (a_{h1}^1, a_{h2}^1, \dots, a_{hm}^1) \in W \subseteq K^m$$

$$u \in (x_1, \dots, x_m) \in K^m$$

$$u \in W \iff \text{rango} \begin{pmatrix} a_{11}^1 & \dots & a_{1h}^1 & x_1 \\ a_{21}^1 & \dots & a_{2h}^1 & x_2 \\ \vdots & & \vdots & \vdots \\ a_{h1}^1 & \dots & a_{hh}^1 & x_h \\ a_{11}^m & \dots & a_{1h}^m & x_{m-h+1} \\ \vdots & & \vdots & \vdots \\ a_{h1}^m & \dots & a_{hh}^m & x_m \end{pmatrix} = h \iff$$

$B = \underbrace{\begin{pmatrix} a_{11}^1 & \dots & a_{1h}^1 \\ \vdots & & \vdots \\ a_{h1}^1 & \dots & a_{hh}^1 \end{pmatrix}}_{\text{rango } h}$

$\iff$

PRIMO MODO (riduzione di Gauss):

una matrice $\bar{B}$ ridotta a gradini ottenuta a partire da B ha h pivot

$$\bar{B} = \begin{pmatrix} p_1^1 & \dots & \dots & b_{11}^1 x_1 + \dots + b_{m1}^1 x_m \\ 0 & p_2^1 & \dots & b_{12}^1 x_1 + \dots + b_{m2}^1 x_m \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \dots & p_h^1 & b_{1h}^1 x_1 + \dots + b_{mh}^1 x_m \\ \vdots & \vdots & \vdots & 0 & \vdots \\ 0 & 0 & 0 & 0 & b_{1h+1}^1 x_1 + \dots + b_{mh+1}^1 x_m \\ 0 & 0 & 0 & 0 & b_{1h+2}^1 x_1 + \dots + b_{mh+2}^1 x_m \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & b_{1m}^1 x_1 + \dots + b_{hm}^1 x_m \end{pmatrix}$$

$$\text{rango}(\bar{B}) = h$$

$$\# \text{ pivot di } \bar{B} = h \iff$$

$$\iff \Sigma_0 \begin{cases} b_{1h+1}^1 x_1 + \dots + b_{mh+1}^1 x_m = 0 \\ \vdots \\ b_{1h+m}^1 x_1 + \dots + b_{mh+m}^1 x_m = 0 \end{cases}$$

è il sist. lineare che cerchiamo
 $S_0 = W$

$\iff$

SECONDO MODO (teorema degli orlati):

$$\text{rango} \begin{pmatrix} a_{11}^1 & \dots & a_{1h}^1 & x_1 \\ a_{21}^1 & \dots & a_{2h}^1 & x_2 \\ \vdots & & \vdots & \vdots \\ a_{h1}^1 & \dots & a_{hh}^1 & x_h \\ a_{11}^m & \dots & a_{1h}^m & x_{m-h+1} \\ \vdots & & \vdots & \vdots \\ a_{h1}^m & \dots & a_{hh}^m & x_m \end{pmatrix} = h$$

$B = \underbrace{\begin{pmatrix} a_{11}^1 & \dots & a_{1h}^1 \\ \vdots & & \vdots \\ a_{h1}^1 & \dots & a_{hh}^1 \end{pmatrix}}_{\text{rango } h}$

Per il Teorema degli orlati; esiste un minore M di B di ordine h con determinante non nullo.

A meno di uno scambio di righe, possiamo supporre che

$$M = \begin{pmatrix} a_{11}^1 & \dots & a_{1h}^1 \\ a_{11}^2 & \dots & a_{1h}^2 \\ \vdots & & \vdots \\ a_{11}^h & \dots & a_{1h}^h \end{pmatrix} \quad |M| \neq 0$$

primi h righe

Allora, $\text{rang}(B) = h \iff$ tutti gli $m-h$ orlati di M hanno determinante nullo

$$M_1, \dots, M_{m-h}$$

$$\Sigma_0: \begin{cases} |M_1| = 0 \\ \vdots \\ |M_{m-h}| = 0 \end{cases} \quad \mathcal{S}_0 = \mathcal{W}. \quad \square$$

Def. Si intende Σ_0 ottenuto nella dimostrazione del precedente teorema si dice RAPPRESENTAZIONE (CARTESIANA) di $\mathcal{W}$.

Esempio:

$$\mathcal{W} \subseteq \mathbb{R}^3 \quad \mathcal{W} = \mathcal{L}((2, 1, -2)) \quad \dim \mathcal{W} = 1$$

$$u = (x_1, x_2, x_3) \in \mathbb{R}^3, \quad u \in \mathcal{W} \iff \text{rang} \begin{pmatrix} \boxed{2} & x_1 \\ 1 & x_2 \\ -2 & x_3 \end{pmatrix} = 1$$

$$\text{PRIMO MODO:} \quad \begin{aligned} \underline{e}^2 &\rightarrow \underline{e}^2 - \frac{1}{2} \underline{e}^1 \\ \underline{e}^3 &\rightarrow \underline{e}^3 + \underline{e}^1 \end{aligned} \quad \begin{pmatrix} 2 & x_1 \\ 0 & x_2 - \frac{1}{2} x_1 \\ 0 & x_3 + x_1 \end{pmatrix}$$

$$\mathcal{W}: \begin{cases} x_2 - \frac{1}{2} x_1 = 0 \\ x_3 + x_1 = 0 \end{cases}$$

$$\text{SECONDO MODO:} \quad M = (2) \quad M_1 = \begin{pmatrix} 2 & x_1 \\ 1 & x_2 \end{pmatrix} \quad M_2 = \begin{pmatrix} 2 & x_1 \\ -2 & x_3 \end{pmatrix}$$

$$|M_1| = 2x_2 - x_1 \quad |M_2| = 2x_3 + 2x_1$$

$$\mathcal{W}: \begin{cases} 2x_2 - x_1 = 0 \\ 2x_3 + 2x_1 = 0 \end{cases}$$

$$\text{Esercizio:} \quad U \subseteq \mathbb{R}^3 \quad U = \mathcal{L}((1, 0, 2), (2, 1, 3))$$

$$\begin{pmatrix} 1 & 2 & x_1 \\ 0 & 1 & x_2 \\ 2 & 3 & x_3 \end{pmatrix} \quad \underline{e}^3 \rightarrow \underline{e}^3 - 2 \underline{e}^1 \quad \begin{pmatrix} 1 & 2 & x_1 \\ 0 & 1 & x_2 \\ 0 & -1 & x_3 - 2x_1 \end{pmatrix} \quad \underline{e}^3 \rightarrow \underline{e}^3 + \underline{e}^2$$

$$\begin{pmatrix} 1 & 2 & x_1 \\ 0 & 1 & x_2 \\ 0 & 0 & x_3 - 2x_4 + x_2 \end{pmatrix}$$

$$U: \begin{cases} x_3 - 2x_4 + x_2 = 0 \end{cases}$$

Applicare anche il SECONDO MODO.

Esercizio $W \subseteq \mathbb{R}^4$

$$W = \mathcal{L}((1, 0, 1, 2), (2, 1, 2, 3), (1, 1, 1, 2))$$

$$\begin{pmatrix} 1 & 2 & 1 & x_1 \\ 0 & 1 & 1 & x_2 \\ 1 & 2 & 1 & x_3 \\ 1 & 3 & 2 & x_4 \end{pmatrix}$$

$$\begin{aligned} \underline{e^3} &\rightarrow \underline{e^3} - \underline{e^1} \\ \underline{e^4} &\rightarrow \underline{e^4} - \underline{e^1} \end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & x_1 \\ 0 & 1 & 1 & x_2 \\ 0 & 0 & 0 & x_3 - x_1 \\ 0 & 1 & 1 & x_4 - x_1 \end{pmatrix}$$

$$\underline{e^4} \rightarrow \underline{e^4} - \underline{e^2}$$

$$\begin{pmatrix} 1 & 2 & 1 & x_1 \\ 0 & 1 & 1 & x_2 \\ 0 & 0 & 0 & x_3 - x_1 \\ 0 & 0 & 0 & x_4 - x_1 - x_2 \end{pmatrix}$$

$$W: \begin{cases} x_3 - x_1 = 0 \\ x_4 - x_1 - x_2 = 0 \end{cases}$$

$$\dim U = 2$$

Esercizio

$$W \subseteq \mathbb{R}^5$$

$$W = \mathcal{L}((2, 1, 0, 3, 1), (2, -1, 1, 4, 0))$$

$$\dim W = 2$$

PRIMO
MODO

$$\begin{pmatrix} 2 & 2 & x_1 \\ 1 & -1 & x_2 \\ 0 & 1 & x_3 \\ 3 & 4 & x_4 \\ 1 & 0 & x_5 \end{pmatrix}$$

$$\begin{aligned} \underline{e^2} &\rightarrow \underline{e^2} - \frac{1}{2} \underline{e^1} \\ \underline{e^4} &\rightarrow \underline{e^4} - \frac{3}{2} \underline{e^1} \\ \underline{e^5} &\rightarrow \underline{e^5} - \frac{1}{2} \underline{e^1} \end{aligned}$$

$$\begin{pmatrix} 2 & 2 & x_1 \\ 0 & -2 & x_2 - \frac{1}{2} x_1 \\ 0 & 1 & x_3 \\ 0 & 1 & x_4 - \frac{3}{2} x_1 \\ 0 & -1 & x_5 - \frac{1}{2} x_1 \end{pmatrix}$$

$$\underline{e^3} \rightarrow \underline{e^3} + \frac{1}{2} \underline{e^2}$$

$$\underline{e^4} \rightarrow \underline{e^4} + \frac{1}{2} \underline{e^2}$$

$$\underline{e^5} \rightarrow \underline{e^5} - \frac{1}{2} \underline{e^2}$$

$$\begin{pmatrix} 2 & 2 & x_1 \\ 0 & -2 & x_2 - \frac{1}{2} x_1 \\ 0 & 0 & \begin{cases} x_3 + \frac{1}{2} x_2 - \frac{1}{4} x_1 \\ x_4 - \frac{3}{2} x_1 + \frac{1}{2} x_2 - \frac{1}{4} x_1 \\ x_5 - \frac{1}{2} x_1 - \frac{1}{2} x_2 + \frac{1}{4} x_1 \end{cases} \\ 0 & 0 & \end{pmatrix}$$

$$W: \begin{cases} -\frac{1}{4} x_1 + \frac{1}{2} x_2 + x_3 = 0 \\ -\frac{3}{4} x_1 + \frac{1}{2} x_2 + x_4 = 0 \\ -\frac{1}{4} x_1 - \frac{1}{2} x_2 + x_5 = 0 \end{cases}$$

SECONDO
MODO

$$\text{rank} \begin{pmatrix} 2 & 1 & 0 & 3 & 1 \\ 2 & -1 & 1 & 4 & 0 \\ x_1 & x_2 & x_3 & x_4 & x_5 \end{pmatrix} = 2$$

$$M = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix} \Rightarrow |M| = 1 \neq 0$$

$$M_1 = \begin{pmatrix} 2 & 1 & 0 \\ 2 & -1 & 1 \\ x_1 & x_2 & x_3 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 1 & 0 & 3 \\ -1 & 1 & 4 \\ x_2 & x_3 & x_4 \end{pmatrix}$$

$$M_3 = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \\ x_3 & x_4 & x_5 \end{pmatrix}$$

$$W: \begin{cases} |M_1| = -2x_3 + x_4 - 2x_2 - 2x_3 = 0 \\ |M_2| = x_4 - 3x_3 - 3x_2 - 4x_3 = 0 \\ |M_3| = x_5 - x_3 - x_2 = 0 \end{cases}$$

$(V, +, \cdot)$ K , $\dim V = n$, $B = (e_1, \dots, e_n)$ basi ordinata di V

$W \subseteq V$ sottosp. vet., $\dim W = h$

$$\phi_B: V \longrightarrow K^n$$

$$u \longmapsto (x_1, \dots, x_n) : u = x_1 e_1 + \dots + x_n e_n$$

$$\phi_B(W) \subseteq K^n, \quad \dim(\phi_B(W)) = \dim W = h$$

sottosp. vet.

Possiamo rappresentare $\phi_B(W): \Sigma_0: AX = 0$

RAPPRESENTAZIONE CARTESIANA
di $\phi_B(W)$

n dice RAPPRESENTAZIONE CARTESIANA
di W rispetto alla basi ordinata B
(riferimento)

Esempio: $V = \mathbb{R}[x] \leq 2$ $B = (1, x, x^2)$

$W = \mathcal{L}(1-x)$ Rappresenta W in B :

$$\phi_B: V \longrightarrow \mathbb{R}^3$$

$$a_0 + a_1 x + a_2 x^2 \longmapsto (a_0, a_1, a_2)$$

$$\phi_B(W) = \mathcal{L}(\phi_B(1-x)) = \mathcal{L}((1, -1, 0))$$

$$\begin{pmatrix} \boxed{1} & x_1 \\ -1 & x_2 \\ 0 & x_3 \end{pmatrix} \xrightarrow{x_2 \rightarrow x_2 + x_1} \begin{pmatrix} 1 & x_1 \\ 0 & x_2 + x_1 \\ 0 & x_3 \end{pmatrix}$$

$$W: \begin{cases} x_2 + x_1 = 0 \\ x_3 = 0 \end{cases}$$

$$\mathcal{S}_0 = \phi_B(W)$$

$$W = \phi_B^{-1}(\mathcal{S}_0)$$

Col teorema degli orbitali:

$$M = (1) \quad |M| = 1$$

$$M_1 = \begin{pmatrix} 1 & x_1 \\ -1 & x_2 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & x_1 \\ 0 & x_3 \end{pmatrix}$$

$$\begin{cases} |M_2| = x_2 + x_4 = 0 \\ |M_3| = x_3 = 0 \end{cases}$$

Rappresentazioni parametriche:

$W \subseteq K^m$ sottosp. vett. numerico, $\dim W = h$

Come nelle dimostrazioni del teorema:

$$\begin{aligned} B = (w_1, \dots, w_h), \quad w_1 = (a_1^1, a_1^2, \dots, a_1^m) \in W \subseteq K^m \\ \text{base di } W \quad \vdots \quad w_h = (a_h^1, a_h^2, \dots, a_h^m) \in W \subseteq K^m \end{aligned} \quad W = \mathcal{L}(B)$$

$$u = (x_1, \dots, x_m) \in K^m$$

$$u \in W \Leftrightarrow u \in \mathcal{L}(w_1, \dots, w_h) \Leftrightarrow \exists t_1, \dots, t_h \in K:$$

$$\begin{aligned} (x_1, \dots, x_m) &= t_1 (a_1^1, a_1^2, \dots, a_1^m) + \dots + t_h (a_h^1, a_h^2, \dots, a_h^m) = \\ &= (a_1^1 t_1 + \dots + a_h^1 t_h, a_1^2 t_1 + \dots + a_h^2 t_h, \dots, a_1^m t_1 + \dots + a_h^m t_h) \\ \begin{cases} x_1 = a_1^1 t_1 + \dots + a_h^1 t_h \\ x_2 = a_1^2 t_1 + \dots + a_h^2 t_h \\ \vdots \\ x_m = a_1^m t_1 + \dots + a_h^m t_h \end{cases} \end{aligned} \quad \text{RAPPRESENTAZIONE PARAMETRICA di } W.$$

Esempio: $U = \mathcal{L}((1, 0, 2), (2, 1, 3)) \subseteq \mathbb{R}^3$

$$\begin{cases} x_1 = t_1 + 2t_2 \\ x_2 = t_2 \\ x_3 = 2t_1 + 3t_2 \end{cases}$$

Esempio: $V = \mathbb{R}[x] \leq 2 \quad B = (1, x, x^2)$
 $W = \mathcal{L}(1-x) \quad \phi_B(1-x) = (1, -1, 0)$

$$\begin{cases} x_1 = t \\ x_2 = -t \\ x_3 = 0 \end{cases} \quad \begin{array}{l} \text{RAPPRESENTAZIONE PARAMETRICA di } W \\ \text{rispetto a } B \\ (\text{in } B) \end{array}$$

OSS

↙ "elimineremo il parametro"

$$\begin{cases} t = x_1 \\ x_2 = -x_1 \\ x_3 = 0 \end{cases} \quad \leftarrow \text{RAPP. CARTESIANA (non parametrica)}$$

VICEVERSA:

$$W: \begin{cases} -\frac{1}{2}x_1 + \frac{1}{2}x_2 + x_3 = 0 \\ -\frac{3}{2}x_1 + \frac{1}{2}x_2 + x_4 = 0 \\ -\frac{1}{2}x_1 - \frac{1}{2}x_2 + x_5 = 0 \end{cases}$$

$$| -\frac{1}{4}x_1 - \frac{1}{2}x_2 + x_5 = 0$$

$$\begin{cases} -x_1 + 2x_2 + 4x_3 = 0 \\ -7x_1 + 2x_2 + 4x_4 = 0 \\ -x_1 - 2x_2 + 4x_5 = 0 \end{cases}$$

$$\begin{pmatrix} -1 & 2 & 4 & 0 & 0 \\ -7 & 2 & 0 & 4 & 0 \\ -1 & -2 & 0 & 0 & 4 \end{pmatrix}$$

$$\begin{cases} x_3 = \frac{1}{4}x_1 - \frac{1}{2}x_2 \\ x_4 = \frac{7}{4}x_1 - \frac{1}{2}x_2 \\ x_5 = \frac{1}{4}x_1 + \frac{1}{2}x_2 \end{cases}$$

$$S_0 = \left\{ (x_1, x_2, \frac{1}{4}x_1 - \frac{1}{2}x_2, \frac{7}{4}x_1 - \frac{1}{2}x_2, \frac{1}{4}x_1 + \frac{1}{2}x_2) \mid x_1, x_2 \in \mathbb{R} \right\}$$

$$\begin{cases} x_1 = t_1 \\ x_2 = t_2 \\ x_3 = \frac{1}{4}t_1 - \frac{1}{2}t_2 \\ x_4 = \frac{7}{4}t_1 - \frac{1}{2}t_2 \\ x_5 = \frac{1}{4}t_1 + \frac{1}{2}t_2 \end{cases} \quad t_1, t_2 \in \mathbb{R}$$

Esercizio:

$$W = \mathcal{L}((1, 2, 0, 2), (0, 0, 1, 2)) \subseteq \mathbb{R}^4$$

$$U = \mathcal{L}((1, 0, 1, 2), (1, 2, 1, 3))$$

Determinare $W \cap U$.

$$W: AX = 0,$$

$$U: A'X = 0$$

$$U \cap W: \begin{cases} AX = 0 \\ A'X = 0 \end{cases}$$

$$\begin{pmatrix} 1 & 0 & x_1 \\ 2 & 0 & x_2 \\ 0 & 1 & x_3 \\ 1 & 2 & x_4 \end{pmatrix} \begin{array}{l} \underline{2} \rightarrow \underline{2} - \underline{1} \\ \underline{4} \rightarrow \underline{4} - \underline{1} \end{array} \begin{pmatrix} 1 & 0 & x_1 \\ 0 & 0 & x_2 - 2x_1 \\ 0 & 1 & x_3 \\ 0 & 2 & x_4 - x_1 \end{pmatrix} \begin{array}{l} \underline{2} \rightarrow \underline{2} \\ \underline{4} \rightarrow \underline{4} - \underline{2} \end{array}$$

$$\begin{pmatrix} 1 & 0 & x_1 \\ 0 & 1 & x_3 \\ 0 & 0 & x_2 - 2x_1 \\ 0 & 2 & x_4 - x_1 \end{pmatrix} \begin{array}{l} \underline{4} \rightarrow \underline{4} - \underline{2} \end{array} \begin{pmatrix} 1 & 0 & x_1 \\ 0 & 1 & x_3 \\ 0 & 0 & x_2 - 2x_1 \\ 0 & 0 & x_4 - x_1 - 2x_3 \end{pmatrix}$$

$$W: \begin{cases} x_2 - 2x_1 = 0 \\ x_4 - x_1 - 2x_3 = 0 \end{cases}$$

$$\begin{pmatrix} \boxed{1} & 0 & 1 & 1 \\ 1 & 2 & 1 & 3 \\ x_1 & x_2 & x_3 & x_4 \end{pmatrix}$$

$$M = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} \quad |M| = 2 \neq 0$$

$$M_1 = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 1 \\ x_1 & x_2 & x_3 \end{pmatrix}$$

$$M_2 = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & 3 \\ x_1 & x_2 & x_4 \end{pmatrix}$$

$$\cup \begin{cases} |M_1| = 2x_3 + x_2 - 2x_1 - x_1 = 0 \\ |M_2| = 2x_4 + x_2 - 2x_1 - 3x_1 = -2x_1 - 2x_1 + 2x_4 = 0 \end{cases}$$

$$W \cap U: \begin{cases} x_2 - 2x_1 = 0 \\ x_4 - x_1 - 2x_3 = 0 \\ 2x_3 - 2x_4 = 0 \\ -2x_1 - 2x_2 + 2x_4 = 0 \end{cases} \quad W$$

$$W: \begin{cases} x_2 - 2x_1 = 0 \\ x_4 - x_1 - 2x_3 = 0 \end{cases}$$

OPPURE:

$$U: \begin{cases} x_1 = t_1 + t_2 \\ x_2 = 2t_2 \\ x_3 = t_1 + t_2 \\ x_4 = t_1 + 3t_2 \end{cases}$$

$$u \in U \iff \exists t_1, t_2 \in \mathbb{R} : u = (t_1 + t_2, 2t_2, t_1 + t_2, \underline{t_1 + 3t_2})$$

$$\mu \in U \cap W \iff \begin{cases} 2t_2 - 2(t_1 + t_2) = 0 \\ t_1 + 3t_2 - t_1 - t_2 - 2(t_1 + t_2) = 0 \end{cases}$$

$$\begin{cases} 2\cancel{t_1} - 2\cancel{t_1} - 2\cancel{t_2} = 0 & \Rightarrow t_2 = 0 \\ 3\cancel{t_2} - \cancel{t_2} - 2\cancel{t_2} = 0 & \Rightarrow 0 = 0 \end{cases}$$

$$u \in U \cap W \Rightarrow \exists t_2 \in \mathbb{R}: u = \underbrace{(t_2, 2t_2, t_2, 3t_2)}_{t_2(1, 2, 1, 3)} \in \mathcal{L}((1, 2, 1, 3))$$